

Munjung Kim

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Education & Training	University of Virginia <i>Doctor of Philosophy, Data Science</i> Advisor: Dr. Yong-Yeol Ahn	Aug 2025 - Present
	Indiana University Bloomington <i>Doctor of Philosophy program, Informatics (transferred with advisor)</i> Advisor: Dr. Yong-Yeol Ahn	Aug 2022 - Aug 2025
	Pohang University of Science and Technology <i>Bachelor of Science, Physics</i>	Feb 2017 - Feb 2022
Work Experience	Nokia Bell Labs Research Intern Identifying job tasks expected to be impacted by disruptive AI technologies using LLMs, text embeddings, and patent citation network analysis [W2]	<i>Cambridge, UK</i> June 2024– Aug 2024
	SK Hynix Quality Control Intern	<i>Icheon, South Korea</i> Jan 2020– Feb 2020
Research Interest	Machine Learning (graph representation learning, language models) Computational Social Science (Innovation, Collective intelligence, Science of science, AI-agent behavior)	
Working Papers	[W1] Munjung Kim and Yong-Yeol Ahn. <i>Mitigating Imitative Falsehoods in LLMs Through Crowdsourcing Techniques</i>. [W2] Munjung Kim, Marios Constantinides, Sanja epanovi, Yong-Yeol Ahn, and Daniele Quercia. <i>Augmenting Minds or Automating Routines? The Potential Impact of Disruptive AI Innovations on U.S. Occupations</i>. [W3] Munjung Kim, Nadav Kunievsky, Robert Ward, Seolmin Yang, and Yong-Yeol Ahn. <i>Murmuration of Scientists</i>. 2026. [W4] Munjung Kim, Jisung Yoon, and Jaehyuk Park. <i>Global Landscape of AI Policy</i>. Jaehyuk Park: corresponding author. 2026.	
Publications	[A1] Munjung Kim, Sadamori Kojaku, and Yong-Yeol Ahn. “Uncovering simultaneous breakthroughs with a robust measure of disruptiveness”. <i>Science Advances</i> 12 (2026). [code] [pip] [blog], eadx3420. doi: 10.1126/sciadv.adx3420. [A2] Rachith Aiyappa, Xin Wang, Munjung Kim, Ozgur Seckin, Yong-Yeol Ahn, and Sadamori Kojaku. “Implicit Degree Bias in the Link Prediction Task”. <i>Proceedings of the 42nd International Conference on Machine Learning (ICML)</i>. Honolulu, Hawaii, 2025. [A3] Munjung Kim, Turulcan Elmas, and Filippo Menczer. “Toxic Synergy Between Hate Speech and Fake News Exposure”. <i>Proceedings of the 18th International AAAI Conference on Web and Social Media (ICWSM 2024)</i> (2024). [A4] Matthew R DeVerna, Rachith Aiyappa, Wanying Zhao, Manita Pote, Bao Tran Truong, David Axelrod, Aria Pessianzadeh, Zoher Kachwala, Munjung Kim, Ozgur Can Seckin, et al. “A multi-platform collection of social media posts about the 2022 US midterm elections”. <i>Proceedings of the international AAAI conference on web and social media</i> (2023).	

- [A5] **Munjung Kim**, Jisung Yoon, Woo-Sung Jung, and Hyunuk Kim. “Quantifying the topic disparity of scientific articles”. *Companion Proceedings of the Web Conference 2022* (2022), pp. 769–773.

Oral

Presentations

Invited Talks

- KDI School Special Lecture 12/17/2024
- KAIST STP Science of Science Research Seminar 12/18/2024
- SNU Business School Workshop 01/03/2025
- KISTI 01/24/2025
- UVA SDS Advisory Board Meeting 04/17/2026
- *Panelist*, Understanding and Scoping Future Research on Disruptive Innovation: A Workshop, National Academies of Sciences, Washington, DC 05/20/2026

Contributed Talks

Munjung Kim and Yong-Yeol Ahn. “Murmuration of Scientists”. NetSci 2026. 2026.

Munjung Kim, Sadamori Kojaku, and Yong-Yeol Ahn. “A Continuous Embedding Measure of Disruptiveness Reveals Hidden Simultaneous Disruption”. IC2S2. 2024.

Munjung Kim, Sadamori Kojaku, and Yong-Yeol Ahn. “Quantifying disruptiveness using neural embedding method”. Netsci 2023. 2023.

Munjung Kim, Jisung Yoon, Hyunuk Kim, and Woo-Sung Jung. “Neural embedding of research papers reveals characteristics of policy research institutes”. Networks 2021: A Joint Sunbelt & NetSci Conference. 2021.

Munjung Kim, Jisung Yoon, and Woo-Sung Jung. “Quantitative Analysis on Institutions’ Activity Policy Research with Embedding Method”. Korea Academy of Complexity Studies. 2019.

Honors

&Awards

2025 December Graduate Conference, Institute for Humane Studies,

2025/ ca. \$100 **Humane Studies Fellowship**, Institute for Humane Studies,

2024/ ca. \$5,000

Complex Systems Summer School, Santa Fe Institute,

2023

National Scholarship for Excellence (Sci. & Eng.), Korea Student Aid Foundation,

2017-2021 / ca. \$20,000

Global Talent Attraction Program (Postponed due to Covid19), Indiana University,

2020 / ca. \$4,000

Semester of High Honor, 2019, 2020, 2021

Senior Mentoring Program Scholarship, 2019,2020

Teaching

Tutorial Instructor, SICSS Korea

06/2026

- “Claude Code in Research for Beginners” (introductory tutorial) [[materials](#)]

Skills

Computer Language: Python, R, Matlab, C, C++, Cython

Workflow for Reproducible Data Analysis: Snakemake

Analysis Modules: Pytorch, Sci-kit Learn, Numpy, Scipy, Pandas, igraph, NetworkX

Languages: English (Proficient), Korean (Native)

Service

Journal Reviewer

- 2025 Nature Matters Arising
- 2025 Network Science
- 2025 npj Complexity

Conference Reviewer

- International Conference on Computational Social Science 2025
- International Conference on Computational Social Science 2024
- International Society for Scientometrics and Informetrics 2023 conference
- International Workshop on Cyber Social Threats (CySoc 23)
- International Workshop on Cyber Social Threats (CySoc 24)